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Examining the Effects of Task Type, Task Order, and Their Interaction on L2 Writing Performance: Insights From Independent and Integrated

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Abstract

This study investigates the effects of task type, task order, and their interaction on second language (L2) writing performance, focusing on independent and integrated writing tasks. While task type has been widely studied, the role of task order and their interaction remains underexplored. This research examines how task configurations influence syntactic complexity, accuracy, lexical complexity, and fluency (CALF) in L2 writing. Four hundred L2 learners performed independent and integrated writing tasks in a counterbalanced order at a testing center at a large university in the US. Their essays were analyzed using fine-grained CALF measures. Results revealed that task type generally had a stronger and more stable effect on CALF metrics, with learners performing better on the integrated task (INTEG) compared to the independent task (IND). Task type emerged as a significant predictor of syntactic complexity, particularly for MLTU, DC/T, and CN/T, with effects enhanced when task order was considered. While task order alone had less impact, significant interaction effects indicated that task order modulated the influence of task type on syntactic complexity measures except for CN/C. Lexical complexity measures similarly highlighted the task type's stronger effect, though task order occasionally reached significance. Interaction effects were

also observed for accuracy (EFC/C) and fluency (WPM), underscoring the combined influence of task type and task order. Notably, CN/C remained unaffected by either task type, task order, or their interaction. These findings highlight the complexity of L2 writing performance and the need to consider task configurations when interpreting writing outcomes.

초록

본 연구는 독립형 쓰기 과업과 통합형 쓰기 과업을 중심으로, 과업 유형, 과업 순서 및 이들의 상호작용이 제2언어 쓰기 능력에 미치는 영향을 탐구한다. 과업 유형은 폭넓게 연구되어 왔지만, 과업 순서와 이들의 상호작용은 상대적으로 연구가 부족하다. 본 연구는 이러한 과업 구성이 L2 쓰기의 구문 복잡성, 정확성, 어휘 복잡성, 유창성(이하 CALF)에 어떻게 영향을 미치는지 살펴본다. 미국 대학의 한 글쓰기 센터에서 400명의 L2 학습자가 독립형 쓰기 과업과 통합형 쓰기 과업을 수행하였다. 이들의 에세이는 세분화된 CALF 척도를 사용하여 분석되었다. 연구 결과, 과업 유형은 CALF 측정 항목에 전반적으로 더 강력하고 안정적인 영향을 미쳤으며, 학습자들은 독립형 쓰기 과업에 비해 통합형 쓰기 과업에서 더 나은 쓰기 능력을 보였다. 과업 유형은 구문 복잡성의 중요한 예측 변수로 나타났으며, 과업 순서를 고려했을 때 그 효과는 더욱 증폭되었다. 과업 순서만으로는 영향이 적었지만, 과업 유형과의 상호작용을 고려하였을 경우 구문 복잡성 측정 항목에 대한 과업 유형의 영향의 정도가 과업 순서에 의해 영향을 받았음을 보여주었다. 어휘 복잡성과 관련하여 과업 유형이 상대적으로 강한 영향을 끼쳤음이 확인되었지만, 과업 순서 또한 전반적으로 유의미한 영향을 보였다. 에세이의 정확성과 유창성에서도 상호작용 효과가 관찰되었는데, 이는 과업 유형과 과업 순서의 통합적인 영향을 받았음을 보여준다. 그러나 절당 복잡 명사어구(Complex Nominals per Clause, CN/C)는 과업 유형, 과업 순서, 또는 이들의 상호작용에 의해 영향을 받지 않았다. 이러한 연구 결과는 L2 쓰기 능력의 복잡성과 쓰기 결과 해석 시 과업 구성을 고려해야 할 필요성을 강조한다.

1 Introduction

The influence of task type on second language (L2) writing has been widely studied, with a growing body of research showing that different task types can significantly affect L2 writing outcomes (e.g., Abdi Tabari et al. 2025; Alexopoulou et al. 2017; Kim and Pae 2021). In addition, Manchón (2014) and Révész et al. (2017) have contributed extensively to this area, particularly in elucidating how task characteristics influence cognitive and linguistic dimensions of L2 writing. While much of the literature has focused on comparing independent and integrated tasks, the effects of these task types on four key dimensions of L2 writing performance—syntactic complexity, accuracy, lexical complexity, and fluency (CALF) (Barrot and Gabinete 2019)—remain insufficiently explored. Independent tasks, which typically require learners to produce content based solely on their own knowledge, contrast with

integrated tasks that involve synthesizing and integrating information from external sources, such as reading or listening materials (Plakans 2010). These task types engage distinct cognitive and linguistic processes, with integrated tasks generally demanding more advanced synthesis and integration skills (Michel et al. 2020). Despite the growing attention to task type as a factor influencing L2 writing performance, its specific impact on CALF has yet to be adequately examined.

Another important, yet underexplored, variable in task-based L2 writing research is task order. Building on Robinson's (2001) Cognition Hypothesis and his 2010 SSARC model of pedagogic task sequencing (stabilize, simplify, automatize, reconstruct, and complexify), an emerging body of research suggests that the order in which tasks are presented can significantly influence L2 writing performance (see Abdi Tabari et al. 2024a for a recent meta-analysis). However, to the best of our knowledge, the effects of task order—particularly the sequencing of independent and integrated tasks—on CALF in L2 writing have not been examined in the literature. Furthermore, the interaction between task type and task order has received scant attention, despite its potential to yield critical insights into both the theoretical foundations of task-based language teaching (TBLT) and its practical application in L2 writing instruction (Ellis 2018). To address these identified gaps in the literature, the present study seeks to:

1. Investigate the extent to which task type influences L2 writing performance in terms of CALF measures.
2. Examine the impact of task order on L2 writing performance, specifically with respect to CALF.
3. Explore the interaction between task type and task order, and compare these findings with the independent effects of each variable.

By examining the effects of task type and task order on fine-grained CALF measures, this study aims to deepen the current understanding of how these task variables, both individually and interactively, affect L2 writing outcomes. This approach aligns with Manchón's call for a more nuanced exploration of the writing outcomes in TBLT (Manchón et al. 2023) and complements Révész's work on how task design can shape L2 writing performance (Révész et al. 2017). The findings will have important implications for the theoretical development of TBLT, as well as practical recommendations for optimizing task design and sequencing in L2 writing classrooms, ultimately contributing to more effective and evidence-based approaches to L2 writing instruction.

2 Literature Review

2.1 Effects of Task Type (Independent Versus Integrated Tasks) on L2 Writing Performance

The distinction between independent and integrated writing tasks has been a central focus in L2 writing research, as these tasks engage different cognitive processes and produce varied linguistic outcomes. Independent tasks typically involve impromptu essay writing in response to a general prompt, drawing on personal experiences within a fixed time frame (Plakans 2008). In contrast, integrated tasks require synthesizing information from multiple source texts into a cohesive written response, demanding that students comprehend, analyze, and integrate various viewpoints (Plakans and Gebril 2013; Weigle and Parker 2012). This process imposes greater cognitive demands, engaging writers in higher-order functions such as analysis, synthesis, and evaluation (Payant et al. 2019). This distinction aligns with Manchón's (2014) argument that task features influence the type and depth of cognitive engagement during L2 writing, thereby shaping linguistic output.

A key distinction between the two task types is the level of cognitive engagement. Integrated tasks, requiring source-based synthesis, demand more complex cognitive activities than independent tasks. These tasks necessitate advanced comprehension, analysis, and the ability to draw meaningful conclusions from multiple sources, in addition to demonstrating writing proficiency. Independent tasks, in contrast, focus more on writing fluency and expression, relying on personal experience or opinion, and do not require the same level of cognitive processing (Plakans 2008).

Several studies have examined the linguistic outcomes of these tasks. Guo et al. (2013) found that integrated tasks led to higher lexical sophistication and greater syntactic complexity than independent tasks. Biber et al. (2014) showed that integrated tasks were linked to more lexical and grammatical features typical of academic writing. Abrams (2019) demonstrated that integrated tasks improved vocabulary use and syntactic complexity, suggesting that synthesizing multiple sources fosters greater linguistic accuracy and variety. Golparvar and Rashidi (2021) identified significant differences, with integrated tasks affecting unit length, coordination, and noun phrase complexity. These findings resonate with Révész et al. (2017), who demonstrated that syntactic and lexical outcomes are significantly shaped by task complexity, especially in the L2 writing context requiring information integration.

Regarding cognitive processes, integrated tasks place less emphasis on planning and translation, as students can rely on provided sources for content and language, leading to greater fluency and more time for monitoring and revision. Plakans (2008) found that

integrated tasks on the TOEFL iBT encouraged rereading the prompt and engaging in task interpretation during writing. These tasks also facilitated planning, likely due to reliance on source material. Independent tasks, however, elicited more initial planning but less during-writing planning and more frequent rereading, suggesting that content generation in these tasks is more cognitively demanding. This reflects Michel et al.'s (Michel et al. 2020) observation that task demands influence the temporal distribution of cognitive effort during writing, shaping how learners plan, monitor, and revise.

Barkaoui (2015) found that participants in independent tasks engaged in more extensive prewriting planning, experienced greater difficulty with content generation, and made more frequent revisions of language use. This suggests that independent tasks, requiring content generation based on personal knowledge, impose a heavier cognitive load, particularly in content creation and language use. Michel et al. (2020) extended this research by examining how writing speed, pausing, eye-gaze behaviors, and associated cognitive processes varied across independent and integrated tasks. Sixty L2 writers performed two independent and two integrated TOEFL iBT writing tasks, with data collected through keystroke logging, eye-movement capture, and stimulated recall interviews. The results showed that, apart from source use in integrated tasks, L2 writers engaged in similar behaviors and cognitive processes across both task types. However, integrated tasks elicited more dynamic and varied behaviors, especially regarding source use, across different stages of the writing process. These behavioral differences echo Révész's (2011) position that writing task complexity and cognitive demands interact dynamically across performance stages, influencing both the quantity and quality of learner output.

In sum, integrated tasks, due to their reliance on source texts, engage learners in higher-level cognitive processes, leading to more sophisticated, academically appropriate language use. Independent tasks, while still cognitively demanding, focus more on content generation and involve distinct patterns of planning and revision. Understanding the effects of task type on writing performance provides valuable insights into task design's role in language learning and writing outcomes. However, a critical gap remains in investigating how task type influences L2 writing performance in terms of CALF measures, which would shed light on the specific impacts of task type on cognitive and linguistic outcomes.

2.2 Theoretical Foundations of Task Complexity and Task Order in L2 Writing Performance

Task complexity in this study was conceptualized within Robinson's Cognition Hypothesis (2001, 2003), which is grounded in a multiple-resource model of attention. This hypothesis posits that attentional resources are not globally limited but can be selectively allocated to

different dimensions of task demands. It further asserts that task complexity is a primary determinant of L2 performance, with distinct complexity features drawing on separate cognitive resources. Within this framework, Robinson (2003) distinguishes between resource-dispersing variables (e.g., planning time, topic familiarity) and resource-directing variables (e.g., number of task elements, reasoning demands). Resource-dispersing variables reduce cognitive load but may divert attention away from form, whereas resource-directing variables simultaneously engage learners in content and linguistic encoding, facilitating greater syntactic and lexical complexity as well as interlanguage development.

In line with this theoretical perspective, the present study manipulated task complexity by comparing two TOEFL iBT-aligned writing tasks: an independent writing task and an integrated writing task. The independent task required participants to construct an argumentative essay based on personal knowledge or opinion, entailing relatively few elements and minimal reasoning demands. The integrated task, by contrast, required synthesizing information from both a reading passage and a lecture, thus increasing the number of information units (± number of elements) and the level of inferential processing (± reasoning demands). These characteristics correspond to key resource-directing dimensions, rendering the integrated task cognitively more complex.

This operationalization aligns with a growing empirical literature demonstrating the multifaceted influence of cognitive task complexity on L2 writing performance. For instance, Ruiz-Funes (2015) reported that higher task complexity enhanced syntactic complexity but negatively affected accuracy and fluency. Similarly, Frear and Bitchener (2015) found that increased cognitive demands reduced the use of complex syntactic structures. Rahimi (2018) and Rahimi and Zhang (2017) further showed that tasks involving greater complexity led to improvements in syntactic subordination, lexical sophistication, content development, and rhetorical organization, albeit with some accuracy trade-offs.

Building on this, Rahimi and Zhang (2019) provided a nuanced exploration of writing task complexity by investigating how increasing the number of elements and reasoning demands affects L2 writing. Their study with sixty upper-intermediate L2 learners showed that task complexity increased syntactic subordination while simultaneously reducing writing accuracy. Similarly, Golparvar and Rashidi (2021) examined the effect of task complexity on integrated writing performance and found that task complexity significantly impacted several measures of syntactic complexity, including mean length of clause, subordination, and mean number of modifiers per noun phrase, as well as lexical diversity and causal cohesion. However, they also reported a negative effect on accuracy and no significant effect on lexical sophistication.

Task sequencing has emerged as a key construct in TBLT research. Robinson's (2010) SSARC model, grounded in the Cognition Hypothesis, advances two core claims: sequencing should be based solely on cognitive factors, and tasks should be ordered from simple to complex. Informed by Wickens' (2002) Multiple Resource Theory, the model emphasizes managing attentional demands across modalities. As Robinson notes, "task sequencing is done by designing and having learners perform tasks simple on all the relevant parameters of task demands first, and then gradually increasing their cognitive complexity on subsequent versions" (2010, 246–247). This approach offers a principled framework for supporting interlanguage development. The SSARC model delineates five pedagogic stages—Stabilize, Simplify, Automatize, Reconstruct, and Complexify—intended to incrementally raise cognitive demands.

This model has informed a growing body of research on how task sequencing influences L2 writing outcomes. However, findings remain inconclusive. Levkina and Gilabert (2014) found that a complex-to-simple sequence facilitated better short-term retention of spatial expressions, while a simple-to-complex order resulted in stronger long-term retention. Lambert and Robinson (2014) similarly reported that learners who followed a simple-to-complex sequence achieved greater long-term gains in task performance, although no significant differences were observed between the SSARC-based and randomized groups.

Allaw and McDonough (2019) investigated the effects of task sequencing in French L2 writing and found that both simple-to-complex and complex-to-simple orders improved performance, but the former led to greater gains in lexical diversity, grammatical accuracy, and overall proficiency. Abdi Tabari and Miller (2021) extended this line of inquiry by examining decision-making tasks in L2 writing, reporting that the simple-to-complex sequence yielded higher syntactic complexity, lexical diversity, accuracy, and fluency. Similarly, Abdi Tabari and Cho (2022) found that the simple-to-complex sequence outperformed the reverse order in terms of syntactic complexity and grammatical accuracy over a short-term period. In a longitudinal study, Abdi Tabari et al. (2024b) compared simple-medium-complex, complex-medium-simple, and randomized sequences across nine weeks. Although all conditions resulted in significant development in CALF, developmental trajectories varied by task order, and no single sequencing pattern proved universally superior.

Despite these contributions, a significant gap remains regarding the differential effects of sequencing independent and integrated writing tasks on L2 written performance. Given their distinct cognitive and linguistic demands, examining how the order of these task types influences CALF measures may yield valuable insights for optimizing task design and sequencing in L2 writing pedagogy.

3 Current Study

The present study seeks to address gaps in the task-based L2 writing literature, specifically concerning the effects of task type (independent versus integrated) and the order in which these tasks are implemented on L2 writing performance. While numerous studies have examined L2 writing products and processes across independent and integrated tasks, there is limited research investigating how task type and task order influence CALF in L2 writing. Furthermore, no prior studies have examined the combined effects of task type and order on L2 writing performance. This study aims to bridge this gap by investigating how independent and integrated tasks, as well as their order of implementation, impact CALF in L2 writing, both individually and synergistically. The study will address the following research questions:

RQ 1. *To what extent does task type affect L2 writing measures of syntactic complexity, accuracy, lexical complexity, and fluency?*

RQ 2. *To what extent does task order affect L2 writing measures of syntactic complexity, accuracy, lexical complexity, and fluency?*

RQ 3. *How do task type, task order, and their interaction affect L2 writing measures of syntactic complexity, accuracy, lexical complexity, and fluency? Furthermore, how do the findings from the previous analyses differ from these findings?*

4 Method

4.1 Participants

The study involved 400 English as a Second Language (ESL) students who were recruited from the ESL writing program at a large university in the United States during the Fall and Spring semesters. The participants represented a diverse group in terms of linguistic and cultural backgrounds, gender, and proficiency levels. Their IELTS proficiency scores ranged from 6.0 to 7.0, with an average of 6.5, corresponding to the C1 level based on the CEFR scale. The participants' ages ranged from 18 to 23 years, with an average age of 20.8 years. Gender distribution was relatively balanced, with 46% identifying as female and 54% as male. The first language (L1) backgrounds of the participants were diverse, including Arabic, Spanish, Mandarin, Hindi, Japanese, Russian, and Portuguese, among others. Arabic speakers constituted the largest group (18%), followed by Spanish speakers (15.25%), Russian (14.5%), Hindi (13.25%), Japanese (13%), Portuguese (12.5%), and Mandarin (13.5%), with the remaining participants distributed among other L1s (see the demographic details in Table 4).

4.2 Writing Tasks

Participants completed two distinct versions of the TOEFL iBT independent and integrated writing tasks, resulting in a dataset of 800 written samples. The task order was counterbalanced across participants to introduce variability in the sequence of task presentation.

For the independent task, participants composed an argumentative essay responding to the prompt: *Do you agree or disagree with the following statement? Students do not respect their teachers as much as they did in the past.* A strict 30-minute time limit was imposed. For the integrated task, participants engaged with materials on the topic of *Living on the Moon*. They began by reading a passage and then listened to a lecture that contradicted the reading. Participants were then asked to write a summary integrating key points from both sources within a 20-minute timeframe, with access to the reading text and their notes. Both tasks were delivered via a simulated TOEFL iBT testing platform, adhering to standard time constraints and conditions. No additional time was provided for planning or rehearsal.

4.3 Procedure

Data collection took place during Week 5 of a 15-week semester. Upon arrival at the university testing center, participants were first provided with informed consent forms. After consenting, participants completed a background questionnaire, which gathered demographic information, language proficiency, and writing experience. This phase took approximately 10 min.

Following the questionnaire, participants engaged in two writing tasks on the same day, presented in a counterbalanced order to minimize order effects. The first task, the independent writing task, required participants to compose an argumentative essay in response to a specific prompt. Participants were given 30 min to complete the task and were instructed to produce a well-organized and coherent argument, focusing on clarity, logical structure, and critical reasoning. The length of the written texts for this task ranged from a minimum of 210 words to a maximum of 500 words, with an average length of 307 words (SD = 80.40).

The second task, the integrated writing task, involved reading a brief text, listening to an audio clip, and then writing a summary or response that integrated information from both sources. Participants were allotted 20 min for this task and were provided with instructions to synthesize the provided materials into a cohesive and relevant response. The length of the written texts for this task ranged from a minimum of 180 words to a maximum of 402 words, with an average of 265 words (SD = 60.34).

Both tasks were conducted in a controlled, quiet environment at the university testing center to ensure consistency across participants. The room was equipped with the necessary resources, including computers with word-processing software, headphones for the listening component of the integrated task, and printed copies of the reading material. Identical instructions and materials were provided to each participant to ensure uniformity in the task conditions. The order of tasks and the timeline were strictly monitored by the first author and research assistants to ensure consistency across sessions. Each session was conducted individually to allow for focused attention, and participants were reminded to adhere to the time limits for each task. Essays were manually checked as part of text pre-processing before conducting the automated analyses.

4.4 Validating Cognitive Task Complexity Manipulations

The present study utilized both independent and integrated writing tasks, which were hypothesized to vary in terms of cognitive demands placed on the learners. To examine the validity of this hypothesis regarding task complexity, a preliminary pilot study was conducted with 100 participants before the study began. Participants first completed the independent writing task, followed by two paper-based post-task questionnaires adapted from Malicka (2018): the Time Judgment Task (TJT) and the Affective Variables Questionnaire (AVQ). Afterward, they completed the integrated writing task and again responded to the same two questionnaires to support the experimental manipulation of cognitive task complexity.

The TJT asked participants to estimate the amount of time they believed they had spent completing each task. These subjective time estimates were later compared to the actual time recorded. The underlying hypothesis was that increased cognitive load would lead to a greater discrepancy between perceived and actual task duration. Meanwhile, the AVQ required participants to rate their experiences across six dimensions using a 9-point Likert scale: mental effort, perceived difficulty, concentration during the task, anxiety levels, confidence in their performance, and overall engagement with the task.

4.5 Data Analysis

We used STATA version 16 as the primary statistical software for data analysis. First, we examined the descriptive statistics and correlations among the eight CALF metrics to understand the details of the collected data. Second, we assessed the homogeneity of the two student groups based on their background information, including age, gender, English proficiency, and the proximity between their L1s and English, through multiple regression analysis. The proximity measure was derived following the rationale proposed by Lee et al. (2022) and Lee and Lee (2024), using a web-based tool for quantifying genetic proximity

(Beaufils & Tomin, 2022). Third, using this background information as control variables, we conducted regression analyses to address the research questions. For the first research question, we performed eight simple regression analyses (one for each CALF metric) to investigate the impact of task type on learners' writing performance. Similarly, for the second research question, we conducted another set of eight simple regression analyses to examine the impact of task order on writing performance. For the third research question, we employed a multiple regression analysis, including task type, task order, and their interaction term as independent variables, to explore how task sequence influences writing performance across the two tasks. Tables and figures were used to present and visualize the findings from these regression models effectively. It is important to note that, as we conducted eight regression models simultaneously for each research question, we applied Bonferroni's correction to adjust the significance level. Accordingly, the threshold for statistical significance was set at $p < 0.006$ (i.e., 0.05/8) to account for multiple comparisons and ensure accurate interpretation of the results (Shadish et al. 2002).

TABLE 1 Descriptive statistics (mean, SD, min, and max) (N = 800).

Variable		Mean	SD	Min	Max
Syntactic complexity	MLTU	16.13	1.61	10.54	19.98
	DC/T	0.68	0.14	0.23	0.99
	CN/T	1.80	0.18	1.09	2.88
	CN/C	1.08	0.24	0.54	1.97
Accuracy	EFC/C	0.60	0.10	0.39	0.99
Lexical complexity	VocD	69.51	7.89	44.80	99.14
	WRDFRQmc	1.73	0.11	1.37	1.99
Fluency	WPM	10.85	1.05	7.12	18.39

5 Results

5.1 Descriptive Statistics and Correlations

Table 1 presents the descriptive statistics for the eight CALF metrics, including the mean, standard deviation, minimum, and maximum values, based on 800 writing samples collected from 400 participants who completed both independent and integrated writing tasks. Key metrics such as Mean Length of T-unit (MLTU), dependent clauses per T-unit (DC/T), and complex nominals per clause (CN/C) reflect variations in syntactic complexity, with MLTU showing a mean of 16.13 and a standard deviation of 1.61. Lexical complexity is represented by metrics like VocD (vocabulary diversity) and word frequency (WRDFRQmc), with VocD

exhibiting moderate variability and an average of 69.51. Fluency, measured by words per minute (WPM), ranges from 7.12 to 18.39, emphasizing individual differences among participants.

Table 2 summarizes the correlations among these metrics, illustrating the relationships between syntactic complexity, accuracy, and lexical complexity. The correlation matrix highlights significant associations; for instance, syntactic complexity (MLTU) positively correlates with accuracy (EFC/C) at $r = 0.30$, $p < 0.001$, and lexical complexity (VocD) shows a positive correlation with fluency (WPM) at $r = 0.13$, $p < 0.001$. Importantly, the correlations between these variables are not excessively high (e.g., $r > 0.8$), which supports the appropriateness of employing multiple measurements. Although these metrics are theoretically interrelated, as they all pertain to writing performance, their relative independence ensures that each captures unique aspects of the construct. This is particularly notable for syntactic and lexical complexity, where multiple distinct metrics were employed. If these metrics had been highly correlated, fewer measures would suffice, improving practicality—similar to accuracy and fluency, which each included only one metric. These findings emphasize the value of a multi-faceted approach in assessing syntactic and lexical complexity within writing.

TABLE 2 Correlation matrix (N = 800).

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1) MLTU	1.00							
(2) DC/T	0.23***	1.00						
(3) CN/T	0.19***	0.30***	1.00					
(4) CN/C	0.09*	0.22***	0.16***	1.00				
(5) EFC/C	0.30***	0.29***	0.13***	0.06	1.00			
(6) Vocd	0.14***	0.04	0.02	−0.02	0.19***	1.00		
(7) WRDFRQmc	0.12**	−0.07	0.05	0.01	0.03	0.15***	1.00	
(8) WPM	0.11**	0.04	0.10**	0.06	0.13***	0.13***	0.11**	1.00

***$p < 0.001$,
**$p < 0.01$,
*$p < 0.05$.

5.2 Validating Cognitive Task Complexity Manipulations

The results from the pilot study clearly supported our hypothesis that the two different writing tasks would vary in terms of cognitive demands placed on the learners. Specifically, we found that the integrated writing task imposed greater cognitive demands on the learners compared to the independent writing task. Across most measures from both the Time Judgment Task (TJT) and the Affective Variables Questionnaire (AVQ), participants reported significantly higher cognitive load for the integrated task. For instance, as shown in Table 3, regarding TJT, participants perceived they had spent significantly more time on the integrated writing task

than they actually had, as reflected in the mean time gap of -149.2 (SD = 138.22) versus 46.2 (SD = 80.26) in the independent writing task. The t-test confirmed this difference was statistically significant ($t = 12.23$, $p < 0.001$).

In the cases of AVQ, participants reported significantly greater mental effort (M = 6.94 versus 3.31), higher perceived task difficulty (M = 7.14 versus 3.4), and elevated anxiety (M = 6.21 versus 3.02) in the integrated writing task, all with statistical significance ($p < 0.001$). Conversely, measures associated with the positive aspect, such as confidence and overall engagement, were significantly higher ($p < 0.001$) for the independent task (confidence: M = 6.61 versus 4.73; engagement: M = 6.55 versus 5.06), again indicating that the integrated task was more cognitively taxing and less comfortable for participants. Notably, the only non-significant difference was in concentration during the task ($t = 0.77$, $p > 0.05$), suggesting that participants were equally focused during both tasks, despite their differing levels of cognitive load.

TABLE 3 Comparison of cognitive load indicators between independent and integrated writing tasks.

Cognitive load indicator		**Independent**		**Integrated**		**T-test**		
		Mean	**SD**	**Mean**	**SD**	**Comparison**	**t-value**	**p**
TJT	Time Gap	46.2	80.26	−149.2	138.22	IND < INTEG	12.23	<0.001
AVQ	Mental Effort	3.31	1.53	6.94	1.2	IND < INTEG	−18.65	<0.001
	Perceived Difficulty	3.4	1.47	7.14	1.06	IND < INTEG	−20.61	<0.001
	Concentration	7.03	1.37	6.86	1.71	IND = INTEG	0.77	>0.05
	Anxiety	3.02	1.63	6.21	1.97	IND < INTEG	−12.48	<0.001
	Confidence	6.61	1.64	4.73	1.76	IND > INTEG	7.81	<0.001
	Engagement	6.55	1.65	5.06	1.32	IND > INTEG	7.06	<0.001

5.3 Group Homogeneity

The two groups with different task orders were evaluated for homogeneity based on background information, including age, gender, English proficiency, and L1 proximity to English. As shown in Table 4, no significant differences were observed between the groups, confirming their comparability on these variables. This ensures that any differences in writing performance can be attributed to task type or order rather than background factors. Nonetheless, these variables were included as control variables in the follow-up regression analysis models to ensure more precise estimation and account for any potential influence.

TABLE 4 Group comparison across background information.

Multiple regression for group comparison			Descriptive statistics			
Independent variable	**B**	**SE**	**IND-to-INT group**		**INT-to-IND group**	
			Mean	**SD**	**Mean**	**SD**
Age	0.01	(0.01)	20.30	(1.38)	20.33	(1.34)
Gender	−0.02	(0.04)	0.49	(0.50)	0.47	(0.50)
English Proficiency	0.005	(0.04)	6.49	(0.42)	6.48	(0.41)
L1-Eng Relation	−0.02	(0.04)	0.42	(0.49)	0.42	(0.49)

5.4 Impact of Task Type on Writing Performance (RQ1)

The results of the simple regression analyses investigating the impact of task type on L2 writing performance are summarized in Table 5. Task type had a significant impact on CALF measures of L2 writing performance, such that learners generally achieved higher CALF scores during integrated writing tasks compared to independent writing tasks, particularly in syntactic complexity—limited to DC/T ($B = 0.03$, $SE = 0.01$, $p < 0.006$) and CN/T ($B = 0.06$, $SE = 0.01$, $p < 0.006$), accuracy ($B = 0.02$, $SE = 0.01$, $p < 0.006$), and lexical complexity (for VocD $B = 1.94$, $SE = 0.55$, $p < 0.006$; for WRDFRQmc $B = 0.03$, $SE = 0.01$, $p < 0.001$). In contrast, we found that learners showed higher fluency in the independent task than the integrated task ($B = -0.24$, $SE = 0.07$, $p < 0.006$). These findings suggest that independent and integrated tasks elicit different levels of syntactic and lexical complexities and accuracy in learners' writing.

TABLE 5 Impact of task type on L2 writing performance.

Eight simple regression models				Inferential statistics			
Writing performance (CALF measurement)		**Task type (INTEG task versus IND task)**		**IND task**		**INTEG task**	
		B	**SE**	**Predicted mean**	**SE**	**Predicted mean**	**SE**
Syntactic complexity	Model 1: MLTU	0.14	(0.11)	16.06*	(0.08)	16.20*	(0.08)
	Model 2: DC/T	0.03*	(0.01)	0.66*	(0.01)	0.69*	(0.01)
	Model 3: CN/T	0.06*	(0.01)	1.77*	(0.01)	1.83*	(0.01)
	Model 4: CN/C	0.01	(0.02)	1.08*	(0.01)	1.08*	(0.01)
Accuracy	Model 5: EFC/C	0.02*	(0.01)	0.58*	(0.00)	0.61*	(0.00)
Lexical complexity	Model 6: VocD	1.94*	(0.55)	68.54*	(0.39)	70.49*	(0.39)
	Model 7: WRDFRQmc	0.03*	(0.01)	1.72*	(0.01)	1.75*	(0.01)
Fluency	Model 8: WPM	−0.24*	(0.07)	10.97*	(0.05)	10.73*	(0.05)

Note: Since eight simple regression models were computed, the significance level was adjusted using Bonferroni's correction (i.e., $p < 0.006$; 0.05/8).
*$p < 0.006$.

5.5 Impact of Task Order on Writing Performance (RQ2)

Table 6 shows the results of the simple regression analyses examining the impact of the task order. Task order had a significant impact on CALF measures of L2 writing performance. Learners in the IND-to-INT Group achieved higher CALF scores during writing tasks compared to those in the INT-to-IND Group, particularly in syntactic complexity—limited to MLTU ($B = -0.79$, $SE = 0.11$, $p < 0.006$), DC/T ($B = -0.06$, $SE = 0.01$, $p < 0.006$), and CN/T ($B = -0.04$, $SE = 0.01$, $p < 0.006$)—and accuracy ($B = -0.04$, $SE = 0.01$, $p < 0.001$). These findings indicate that the sequence in which writing tasks are presented can influence learners' writing performance, particularly in areas related to syntactic complexity and accuracy.

TABLE 6 Impact of task order on L2 writing performance.

Eight simple regression models				**Descriptive statistics**			
Writing performance (CALF measurement)		**Task order (INT-to-IND group versus IND-to-INT group)**		**IND-to-INT group**		**INT-to-IND group**	
		B	**SE**	**Mean**	**SE**	**Mean**	**SE**
Syntactic complexity	Model 1: MLTU	−0.79*	(0.11)	16.52*	(0.08)	15.74*	(0.08)
	Model 2: DC/T	−0.06*	(0.01)	0.71*	(0.01)	0.65*	(0.01)
	Model 3: CN/T	−0.04*	(0.01)	1.82*	(0.01)	1.78*	(0.01)
	Model 4: CN/C	−0.00	(0.02)	1.08*	(0.01)	1.08*	(0.01)
Accuracy	Model 5: EFC/C	−0.04*	(0.01)	0.61*	(0.00)	0.58*	(0.00)
Lexical complexity	Model 6: VocD	−0.86	(0.56)	69.94*	(0.39)	69.08*	(0.39)
	Model 7: WRDFRQmc	−0.00	(0.01)	1.73*	(0.01)	1.73*	(0.01)
Fluency	Model 8: WPM	−0.05	(0.07)	10.87*	(0.05)	10.83*	(0.05)

Note: Since eight simple regression models were computed, the significance level was adjusted using Bonferroni's correction (i.e., $p < 0.006$; 0.05/8).
*$p < 0.006$.

5.6 Combined Effects of Task Type, Task Order, and Interaction (RQ3)

Including both task type and task order, along with their interaction effect, in the multiple regression analyses provided deeper insights into how the impact of task type on learners' L2 writing performance varied depending on task order (Table 7). This finding highlights that the effect of task type is influenced by the order in which tasks are administered. Therefore, jointly considering task type and task order is essential for a nuanced and comprehensive understanding of their impact on writing performance.

In the following, we included a detailed description of the findings. For each CALF metric, we review how the results of simple regressions for task type and task order individually differed from those of the multiple regression analysis incorporating task type, task order, and their interaction (Table 7). Additionally, we describe how the effects of task type varied across task order based on the results in Table 8, which summarizes the predicted means (i.e., the expected values of the outcome variable estimated by the regression model, adjusted for task type, task

order, their interaction, and learners' background information) for the four cases: (1) IND task in the IND-to-INT group, (2) INT task in the IND-to-INT group, (3) IND task in the INT-to-IND group, and (4) INT task in the INT-to-IND group, for each of the eight metrics.

TABLE 7 Combined effects of task type, task order, and interaction on L2 writing performance.

Writing performance (CALF measurement)		RQ#1		RQ#2		RQ#3. multiple regression model					
		Task type (INT task versus IND task)		Task order (INT-to-IND group versus IND-to-INT group)		Task type (INT task versus IND task)		Task order (INT-to-IND group versus IND-to-INT group)		Interaction Effect (task Type × task Order)	
		B	SE	B	SE	B	SE	B	SE	F	*p*
Syntactic Complexity	Model 1: MLTU	0.14	(0.11)	−0.79*	(0.11)	0.75*	(0.15)	−0.18	(0.15)	28.88	<0.006
	Model 2: DC/T	0.03*	(0.01)	−0.06*	(0.01)	0.06*	(0.01)	−0.03	(0.01)	20.49	<0.006
	Model 3: CN/T	0.06*	(0.01)	−0.04*	(0.01)	0.15*	(0.02)	0.04	(0.02)	27.86	<0.006
	Model 4: CN/C	0.01	(0.02)	−0.00	(0.02)	0.03	(0.02)	0.02	(0.02)	0.70	0.55
Accuracy	Model 5: EFC/C	0.02*	(0.01)	−0.04*	(0.01)	0.07*	(0.01)	0.01	(0.01)	26.49	<0.006
Lexical Complexity	Model 6: VocD	1.94*	(0.55)	−0.86	(0.56)	3.39*	(0.78)	0.59	(0.78)	7.26	<0.006
	Model 7: WRDFRQmc	0.03*	(0.01)	−0.00	(0.01)	0.07*	(0.01)	0.04*	(0.01)	13.40	<0.006
Fluency	Model 8: WPM	−0.24*	(0.07)	−0.05	(0.07)	0.45*	(0.10)	0.64*	(0.10)	37.36	<0.006

Note: Since eight simple regression models were computed, the significance level was adjusted using Bonferroni's correction (i.e., $p < 0.006$; 0.05/8).
*$p < 0.006$.

TABLE 8 Predicted means and standard errors of L2 writing performance for task type × task order.

Writing performance (CALF measurement)		IND task × IND-to-INT group		INT task × IND-to-INT group		IND task × INT-to-IND group		INT task × INT-to-IND group	
		Predicted Mean	SE	Predicted Mean	SE	Predicted Mean	SE	Predicted Mean	SE
Syntactic complexity	Model 1: MLTU	16.15*	(0.11)	16.90*	(0.11)	15.97*	(0.11)	15.50*	(0.11)
	Model 2: DC/T	0.68*	(0.01)	0.74*	(0.01)	0.64*	(0.01)	0.65*	(0.01)
	Model 3: CN/T	1.75*	(0.01)	1.90*	(0.01)	1.79*	(0.01)	1.77*	(0.01)
	Model 4: CN/C	1.07*	(0.02)	1.10*	(0.02)	1.09*	(0.02)	1.07*	(0.02)
Accuracy	Model 5: EFC/C	0.58*	(0.01)	0.65*	(0.01)	0.59*	(0.01)	0.57*	(0.01)
Lexical complexity	Model 6: Voc*D*	68.25*	(0.55)	71.64*	(0.55)	68.84*	(0.55)	69.33*	(0.55)
	Model 7: WRDFRQmc	1.70*	(0.01)	1.77*	(0.01)	1.74*	(0.01)	1.73*	(0.01)
Fluency	Model 8: WPM	10.65*	(0.07)	11.10*	(0.07)	11.29*	(0.07)	10.36*	(0.07)

Note: Since eight simple regression models were computed, the significance level was adjusted using Bonferroni's correction (i.e., $p < 0.006$; 0.05/8).
*$p < 0.006$.

We also examine plots (Figure 1) of these predicted means with error bars estimated based on the adjusted significance level ($p < 0.006$). These visualizations help readers intuitively understand the interaction effect. For example, parallel lines in a plot suggest that groups follow a general trend and there is no interaction between variables. In contrast, non-parallel lines (e.g., crossing, diverging, or converging lines) indicate an interaction, where the effect of task type varies depending on task order.

5.6.1 Syntactic Complexity

As shown in Table 7, except for the case of CN/C, the three remaining metrics—MLTU, DC/T, and CN/T—indicated that the results of simple regressions for task type and task order individually differed from those of multiple regression analysis incorporating task type, task order, and their interaction.

First, regarding MLTU, task type alone did not have a significant impact, whereas task order showed a significant effect ($B = -0.79$, $SE = 0.11$, $p < 0.006$). However, when task type and task order were analyzed together, task type had a significant impact ($B = 0.75$, $SE = 0.15$, $p < 0.006$), while task order did not. Although some of these main effects were statistically significant, it is plausible that their influence varied across groups, as indicated by the significant interaction term ($F = 28.88$, $p < 0.006$) and visual cues (i.e., crossing lines) from the left plot in the first row of Figure 1.

The predicted means reported in Table 8 further clarify these results. Learners in the IND-to-INT group performed better on MLTU in their INT task (M = 16.90, $SE = 0.11$) compared to their IND task (M = 16.15, $SE = 0.11$). In contrast, learners in the INT-to-IND group showed no significant difference across task types. Additionally, in the INT task, the IND-to-INT group (M = 16.90, $SE = 0.11$) outperformed the INT-to-IND group (M = 15.50, $SE = 0.11$) in MLTU, whereas no significant differences were observed between the two groups in their IND task.

Second, regarding DC/T, we previously found that task type alone significantly influenced DC/T ($B = 0.03$, $SE = 0.01$, $p = 0.003$), and similarly, task order alone had a significant negative effect ($B = -0.06$, $SE = 0.01$, $p < 0.006$). However, when task type and task order were analyzed together in the multiple regression model, task type's effect increased slightly ($B = 0.06$, $SE = 0.01$, $p < 0.006$), whereas task order's effect was no longer significant ($B = -0.03$, $SE = 0.01$, $p = 0.009$). These changes suggest that task type had a more stable and direct influence on DC/T, while task order's significance, which was shown in the simple regression analysis, was partially mediated when combined with task type in the multiple regression analysis. The significant interaction term ($F = 20.49$, $p < 0.006$) reinforces the idea that their combined effects are noteworthy.

The predicted means in Table 8 provide further clarity, along with the corresponding plot in Figure 1. Regarding the impact of task type, learners in the IND-to-INT group performed significantly better on the INT task (M = 0.74, *SE* = 0.01) than the IND task (M = 0.68, *SE* = 0.01). In contrast, the INT-to-IND group showed marginal differences across task types (M = 0.65, *SE* = 0.01 for INT versus M = 0.64, *SE* = 0.01 for IND). Regarding the impact of task order, although learners in the IND-to-INT group had higher scores in DC/T than those in the INT-to-IND group for both the IND and INT tasks, the difference was not significant, according to the overlapped error bars of the lines in the right plot in the first row of Figure 1.

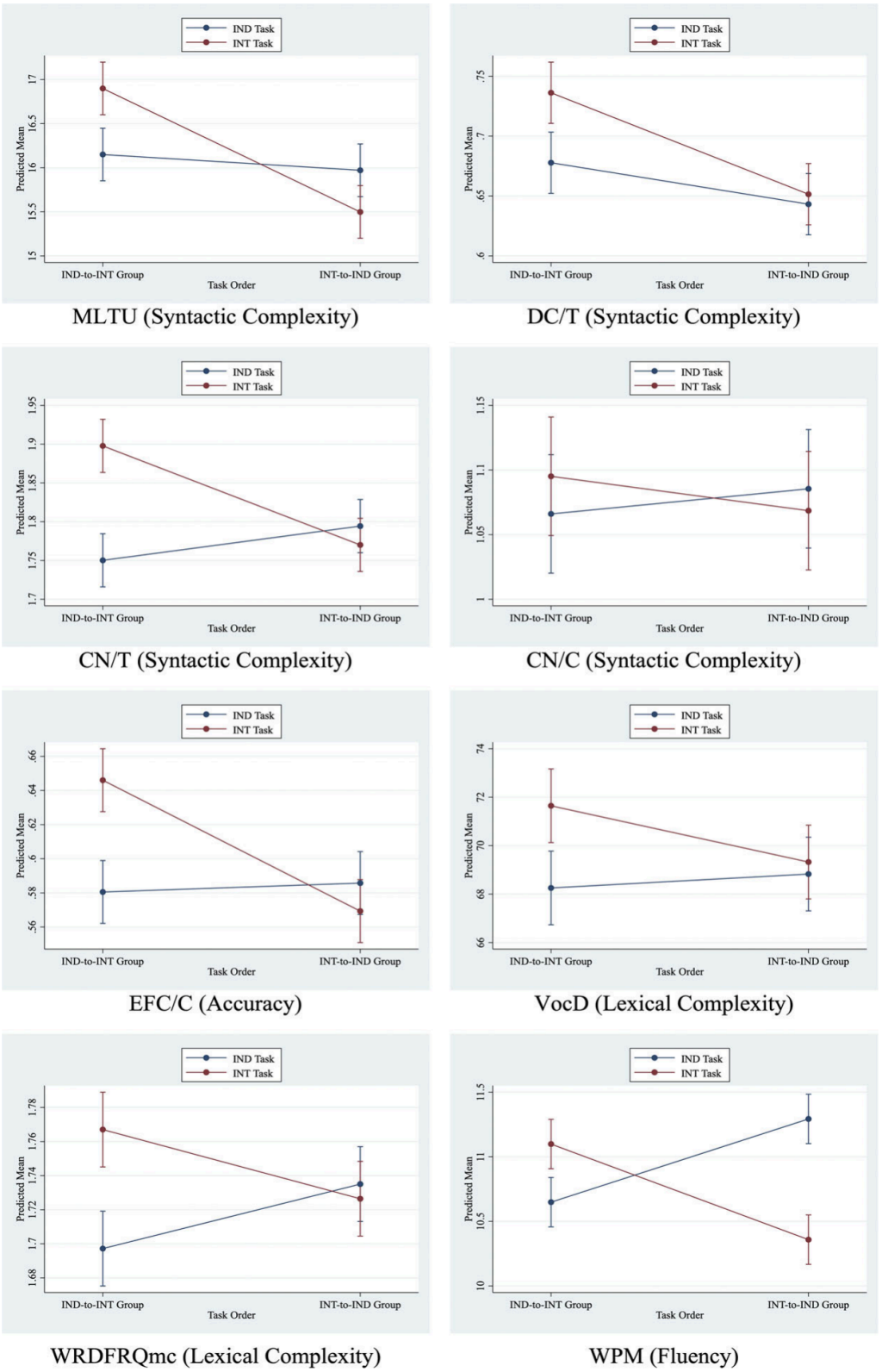


FIGURE 1 Plots for CALF measurement results for task order, task type, and their interaction effect. *Note:* Since eight simple regression models were computed, the significance level for their error bars was adjusted using Bonferroni's correction (i.e., $p < 0.006$; 0.05/8).

Third, regarding CN/T, task type alone had a significant positive effect on CN/T ($B = 0.06$, $SE = 0.01$, $p < 0.001$), and task order alone was significant, too ($B = -0.04$, $SE = 0.01$, $p < 0.006$). However, when task type and task order were analyzed together, the effect of task type increased substantially ($B = 0.15$, $SE = 0.02$, $p < 0.006$), while task order became non-significant ($B = 0.04$, $SE = 0.02$, $p = 0.012$). Moreover, the significant interaction term ($F = 27.86$, $p < 0.006$) indicates that the effect of task type on CN/T varied depending on task order. These results suggest that task order plays a more nuanced role in modulating the effect of task type in the multiple regression analysis compared to the simple regression models.

The predicted means from Table 8 and the corresponding plot in Figure 1 provide further insight. Regarding task type, learners in the IND-to-INT group performed significantly better in the INT task (M = 1.90, $SE = 0.01$) than in the IND task (M = 1.75, $SE = 0.01$). However, learners in the INT-to-IND group displayed slightly higher performance in the IND task (M = 1.79, $SE = 0.01$) compared to the INT task (M = 1.77, $SE = 0.01$), although the difference was not significant. Regarding task order, the IND-to-INT group outperformed the INT-to-IND group on the INT task, while the opposite pattern was observed for the IND task. These differences are reflected in the non-parallel (but crossing) lines in the second-row left plot of Figure 1, emphasizing the significance of the interaction term.

Fourth, regarding CN/C, task type alone did not significantly affect CN/C ($B = 0.01$, $SE = 0.02$, $p = 0.647$), nor did task order alone ($B = -0.00$, $SE = 0.02$, $p = 0.916$). These findings remained consistent when task type and task order were analyzed together in the multiple regression model, where neither task type ($B = 0.03$, $SE = 0.02$, $p = 0.154$) nor task order ($B = 0.02$, $SE = 0.02$, $p = 0.388$) was significant, and the interaction term was also not significant ($F = 0.70$, $p = 0.55$). These results indicate that CN/C remains relatively unaffected by task type, task order, or their interaction, regardless of the analysis model.

Table 8 and Figure 1's plot for CN/C support this interpretation. Learners in the IND-to-INT group achieved slightly higher CN/C scores on the INT task (M = 1.10, SE = 0.02) than on the IND task (M = 1.07, SE = 0.02). For the INT-to-IND group, CN/C scores were marginally better for the IND task (M = 1.09, SE = 0.02) than the INT task (M = 1.07, SE = 0.02). However, the overlapping error bars and flat trend lines in the second-row right plot of Figure 1 confirm that these differences were not statistically meaningful.

5.6.2 Accuracy

For accuracy, based on the results regarding EFC/C, task type alone significantly increased accuracy ($B = 0.02$, $SE = 0.01$, $p = 0.002$), while task order alone significantly decreased it ($B = -0.04$, $SE = 0.01$, $p < 0.006$). When task type and task order were analyzed together, the effect of task type increased ($B = 0.07$, $SE = 0.01$, $p < 0.006$), while the effect of task order

was no longer significant (B = 0.01, SE = 0.01, p = 0.339). The significant interaction term (F = 26.49, p < 0.006) reveals that task order modulated the effect of task type on EFC/C in the multiple regression model, a nuance that was not apparent in the simple regression analyses.

The predicted means in Table 8 clarify these results further. Regarding task type, the IND-to-INT group performed significantly better on the INT task (M = 0.65, SE = 0.01) than the IND task (M = 0.58, SE = 0.01). Conversely, the INT-to-IND group showed slightly higher accuracy on the IND task (M = 0.59, SE = 0.01) compared to the INT task (M = 0.57, SE = 0.01). Regarding task order, the IND-to-INT group outperformed the INT-to-IND group on the INT task, but this difference was not significant for the IND task, as shown by the overlapped error bars in Figure 1. Further, these differences across task type and task order are reflected in the non-parallel (but crossing) lines in the third-row left plot of Figure 1, emphasizing the significance of the interaction term.

5.6.3 Lexical Complexity

As shown in Table 7, the two metrics for lexical complexity—Voc*D* and WRDFRQmc—exhibited slightly different patterns between the results of simple regressions and those of multiple regressions. First, regarding Voc*D*, task type alone significantly influenced lexical diversity (B = 1.94, SE = 0.55, p < 0.006), while task order alone was not significant (B = −0.86, SE = 0.56, p = 0.123). In the multiple regression model, the effect of task type increased (B = 3.39, SE = 0.78, p < 0.006), while task order remained non-significant (B = 0.59, SE = 0.78, p = 0.448). The interaction term reached significance at the adjusted significance level (F = 7.26, p < 0.006), indicating that task order had a significant influence on task type's effect on lexical diversity.

Table 8 and the plot in Figure 1 highlight these trends. Learners in the IND-to-INT group had higher Voc*D* scores on the INT task (M = 71.64, SE = 0.55) than on the IND task (M = 68.25, SE = 0.55). For the INT-to-IND group, Voc*D* remained more stable across tasks, with scores of M = 69.33 (SE = 0.55) for the INT task and M = 68.84 (SE = 0.55) for the IND task. The interaction is visualized in Figure 1's third-row right plot, where the lines converged but did not successfully cross, consistent with the lack of statistical significance for the interaction term.

Second, regarding WRDFRQmc, task type alone significantly increased lexical complexity (B = 0.03, SE = 0.01, p < 0.006), while task order alone was not significant (B = −0.00, SE = 0.01, p = 0.899). In the multiple regression model, task type's effect became stronger (B = 0.07, SE = 0.01, p < 0.006), and task order reached statistical significance (B = 0.04, SE = 0.01, p < 0.006). The significant interaction term (F = 13.40, p < 0.006) suggests that task order influenced task type's effect.

Table 8 shows that learners in the IND-to-INT group achieved higher WRDFRQmc scores on the INT task (M = 1.77, *SE* = 0.01) compared to the IND task (M = 1.70, *SE* = 0.01). In contrast, learners in the INT-to-IND group showed more consistent performance, with scores of M = 1.74 (*SE* = 0.01) in their INT task and M = 1.73 (*SE* = 0.01) in their IND task. Figure 1's third-row left plot illustrates these differences, highlighting the IND-to-INT group's noteworthy improvement in the INT task.

5.6.4 Fluency

For fluency, based on the results regarding WPM, task type alone significantly affected fluency (*B* = −0.24, *SE* = 0.07, $p < 0.006$), whereas task order alone did not (*B* = −0.05, *SE* = 0.07, $p = 0.515$). In the multiple regression model, however, task type remained significant (*B* = 0.45, *SE* = 0.10, $p < 0.006$), and task order also became significant (*B* = 0.64, *SE* = 0.10, $p < 0.006$). The significant interaction term (F = 37.36, $p < 0.006$) suggests that task type and order interactively influenced fluency.

Table 8 reveals that learners in the IND-to-INT group performed better on the INT task (M = 11.10, *SE* = 0.07) than on the IND task (M = 10.65, *SE* = 0.07). In contrast, learners in the INT-to-IND group performed better on the IND task (M = 11.29, *SE* = 0.07) compared to the INT task (M = 10.36, *SE* = 0.07). The interaction is illustrated in Figure 1's bottom right plot, where the two groups' performance trends cross.

5.7 Summary of Key Findings

Overall, the findings highlight that task type and task order have varied influences across the eight CALF metrics, with task type generally demonstrating a stronger and more stable effect, indicating that learners performed higher on CALF metrics in the INTEG task than in the IND task. For syntactic complexity measures (particularly for MLTU, DC/T, and CN/T), task type often emerged as a significant predictor, with its effects enhanced when task order was simultaneously considered. Task order alone was less influential, and its significance often diminished in the presence of task type, as seen in measures like DC/T. However, interaction effects were frequently significant, indicating that task order modulates the influence of task type on performance metrics like MLTU, CN/T, and WRDFRQmc. Lexical complexity measures (Voc*D* and WRDFRQmc) similarly underscored task type's stronger impact, though task order occasionally reached significance in the multiple regression models. Accuracy (EFC/C) and fluency (WPM) also demonstrated significant interaction effects, highlighting the combined influence of task type and task order. Notably, CN/C remained unaffected by either

task type or task order across all analyses. These results underscore the complexity of L2 writing performance and emphasize the need to consider task configuration effects when interpreting writing performance outcomes.

6 Discussion

The study explored the impact of task type (independent versus integrated) and task order (independent-to-integrated versus integrated-to-independent) on L2 writing performance, measured across four key domains: syntactic complexity, accuracy, lexical complexity, and fluency. The findings indicate that both task type and task order, along with their interaction effects, influence L2 learners' writing performance, but with varying degrees of impact on each CALF metric.

6.1 Task Type and Its Effect on the CALF of L2 Writing

Task type emerged as a key determinant influencing L2 writing performance, particularly across the complexity, accuracy, lexis, and fluency (CALF) metrics. The integrated task (INTEG) consistently elicited higher performance than the independent task (IND). Unlike the independent task, the integrated task required learners to generate ideas while synthesizing information from multiple sources, engaging in higher-order cognitive processes such as comparison, synthesis, and argumentation (Plakans and Gebril 2013; Weigle and Parker 2012). These processes, in turn, fostered the production of more complex syntactic structures (Biber et al. 2014), especially when cognitive resources were freed up by content support, as shown in Révész et al. (2017).

The enhanced performance in the integrated task suggests that integrating multiple sources imposes a greater cognitive load, compelling learners to engage in deeper processing. Specifically, synthesizing information from diverse input modalities (e.g., reading and listening) necessitates higher cognitive effort, resulting in more complex written responses (Abrams 2019). Révész et al. (2017) found that when content support was available, learners paused less and revised more, which facilitated attention to linguistic encoding and ultimately yielded higher syntactic complexity. Similarly, Manchón (2014) emphasized the importance of learners' internal interpretation of task demands and their perceived goals, arguing that these factors shape the problem-solving activity during writing and affect processing depth. This learner-task interface likely played a role in how students approached the cognitively demanding integrated task. For instance, the dual processing required in synthesizing content often leads to increased use of subordinate clauses (e.g., higher Dependent Clauses per T-unit, DC/T) and longer sentence structures (e.g., higher Mean Length of T-unit, MLTU) (Golparvar and Rashidi 2021).

This effect was particularly evident in measures of syntactic complexity. Learners produced significantly higher scores in MLTU, DC/T, and Complex Nominals per Clause (CN/T) during the integrated task than the independent task. Prior research corroborates this, showing that tasks requiring the synthesis of multiple inputs lead to the production of more complex sentence structures, as learners establish relationships between disparate ideas (Frear and Bitchener 2015; Plakans 2008). This aligns with Cumming's (2013) view that communicative tasks involving the integration of diverse knowledge sources promote more elaborated language use. Moreover, Manchón et al. (2023) found that while working memory did not significantly affect written performance, L2 proficiency emerged as the most robust predictor across task conditions, reinforcing the idea that cognitive-linguistic resources interact differently depending on learners' developmental readiness and how they interpret task demands.

The integrated task also fostered significantly greater lexical diversity (VocD) and lexical sophistication (WRDFRQmc) compared to the independent task. Engaging with varied input materials required learners to synthesize nuanced information, thereby encouraging the use of a broader range of vocabulary. This process aligns with Meara's (1996) vocabulary knowledge framework, suggesting that tasks involving diverse inputs drive learners to expand their lexical repertoire and select more precise words. Consistent with this, Révész et al. (2017) observed that limited content support led to greater cognitive burden, which in turn reduced lexical sophistication. In contrast, content-supported conditions allowed learners to deploy more sophisticated lexis. Furthermore, as Manchón (2020) notes, writing tasks serve not only as assessment tools but also as meaningful sites for language learning when task engagement involves active problem-solving, self-regulation, and linguistic exploration. In the integrated task, learners' engagement with complex ideas appears to have promoted such cognitive-linguistic development. The independent task, by contrast, provided fewer opportunities to engage with diverse and refined lexical choices (Biber et al. 2014), highlighting the linguistic affordances of integrated task conditions (Payant et al. 2019).

Regarding accuracy (EFC/C) and fluency (WPM), the structured nature of the integrated task also yielded significant improvements. Writing in response to complex materials encouraged learners to focus on clarity and precision, which enhanced accuracy. This finding aligns with studies suggesting that responding to complex input can improve accuracy in L2 writing (Frear and Bitchener 2015; Plakans et al. 2019). Fluency, while improved, showed a less pronounced effect compared to accuracy and syntactic complexity. This is consistent with literature indicating that writing fluency is influenced more by task familiarity and cognitive automaticity than by input complexity (Abdi Tabari et al. 2024c; Barkaoui 2015; Michel et al. 2020). The increased processing demands of the integrated task may have slowed writing

speed, though its structured nature still supported modest gains in fluency. As Manchón (2014) emphasizes, learners' perceived goals and self-regulatory strategies can strongly mediate task performance, potentially explaining the variance in fluency outcomes.

6.2 Task Order and Its Effect on the CALF of L2 Writing

The findings of this study underscore the significant role of task order in influencing L2 writing performance. Learners in the IND-to-INT group significantly outperformed their INT-to-IND counterparts on specific metrics, including syntactic complexity (MLTU, DC/T, and CN/T), and showed a slight advantage in accuracy (EFC/C). However, fluency (WPM) and lexical complexity (Voc*D* and WRDFRQmc) measures revealed relatively balanced performance across the two groups, suggesting that task sequencing has a nuanced impact on different aspects of writing. These results can be explained through Robinson's (2010) SSARC model, as well as considerations of learning transfer and task familiarity.

Starting with the independent task, which imposes less cognitive strain, enables learners to develop and practice foundational writing skills, particularly fluency and accuracy, without the added complexity of integrating information from multiple sources. This initial task serves as a foundation for the more demanding integrated task, which requires synthesizing and organizing information from various texts (Barzilai et al. 2018). Learners in the IND-to-INTEG group, having already mastered basic skills, are better equipped to manage the higher cognitive load of the integrated task. This is likely to result in writing that is more accurate and exhibits greater syntactic complexity. These findings illustrate that task sequencing can influence the CALF of L2 writing, with a gradual increase in cognitive complexity leading to better overall performance on subsequent writing measures (Abdi Tabari and Cho 2022).

The sequencing of tasks from independent to integrated may also facilitate the transfer of writing skills across tasks (Gathercole et al., 2019). Learners in the IND-to-INTEG group likely internalized essential skills, such as structuring ideas and ensuring accuracy, during the independent task. These skills were then transferred to the integrated task, enabling learners to focus on higher-level writing skills, such as syntactic complexity and accuracy. For instance, by the time learners in the IND-to-INTEG group engaged with the integrated task, they were better able to produce a longer mean of T-units (MLTU) and demonstrate higher accuracy due to the cumulative effects of their prior task experience. This transfer of skills contributed to the higher performance observed in this group, especially in specific CALF measures.

Another key factor is task familiarity. Integrated tasks are often more complex, requiring learners to synthesize and coordinate information from multiple sources, which can be overwhelming if encountered first. Presenting the independent task first allows learners to become familiar with the process of writing and organizing their thoughts, reducing the

cognitive demands when transitioning to the integrated task. This familiarity likely contributed to the higher performance in specific CALF measures for the IND-to-INTEG group, as learners had already developed a solid foundation in managing fluency and accuracy before tackling the challenges of the integrated task (Plakans et al. 2019). Moreover, the confidence gained from success in the independent task may have facilitated learners' ability to handle more complex syntactic structures and use a broader range of vocabulary in the integrated task (Cumming et al. 2005).

6.3 Task Type, Task Order, and Their Interaction Effects on the CALF of L2 Writing

The interaction effects between task type and task order were significant across most metrics, except for CN/C. These interactions significantly influenced—and often overshadowed—the independent effects of task type and task order observed in earlier analyses. Specifically, the findings indicate that task type and task order do not operate independently but instead interact in complex ways. For instance, the main effect of task type on all CALF measures (except for CN/C) became more pronounced when analyzed alongside task order and their interaction effects. Conversely, the main effect of task order on syntactic complexity and accuracy diminished, while its influence on lexical complexity (WRDFRQmc) and fluency increased. These results suggest that the sequence in which tasks are performed can amplify or mitigate the effects of task type (Lee 2021), underscoring the importance of the interaction between task types and task orders identified in this study.

For instance, learners in the IND-to-INTEG group likely benefited from having developed content and organizational strategies during the independent task. This foundation enabled them to approach the more cognitively demanding integrated task with enhanced cognitive resources and skills, leading to greater syntactic and lexical complexity (Arnold et al. 2017). This transfer effect aligns with findings from Révész et al. (2017), which showed that reduced cognitive burden, such as that provided through content support, enabled more frequent revisions and greater linguistic complexity, particularly in lexical sophistication. In the current study, a similar scaffolding mechanism appears to be at play, as the initial independent task may have provided conceptual content support that lightened the cognitive load of the integrated task.

Moreover, the task order likely provided a form of cognitive scaffolding: starting with the independent task allowed learners to practice essential writing skills, which were transferable to the integrated task. This scaffolding effect may have contributed to improved L2 writing performance, particularly in areas requiring higher cognitive and linguistic resources, such as syntactic complexity (MLTU, DC/T). As Manchón (2014) argues, learners' internal perceptions of task goals and expected outcomes shape their problem-solving approach during

writing. Thus, the order in which tasks are experienced may affect how learners mentally frame and perform those tasks, potentially boosting performance when the cognitively simple task precedes the cognitively complex one.

Task order is also thought to influence how learners allocate cognitive resources. Integrated tasks require more effort to synthesize information, and learners who begin with independent tasks may experience less cognitive overload when transitioning to integrated tasks (Michel et al. 2020). This aligns with Révész et al.'s (Révész et al. 2017) conclusion that cognitive load affects attention to linguistic encoding; thus, when cognitive demands are managed through strategic sequencing, learners may be better positioned to produce syntactically and lexically richer texts. Conversely, starting with integrated tasks might not offer learners enough time to adjust to the writing process, which could explain why the INTEG-to-IND group performed less effectively on measures like MLTU and DC/T.

When narrowing the scope to each CALF metric, it is noteworthy that the fluency metric demonstrated the largest interaction effect, as evidenced by its highest F-statistic in Table 7 and the distinct crossing lines in Figure 1, where neither end overlapped at the 95% confidence intervals. This was a unique pattern among the CALF metrics, as other metrics showed some degree of overlap at one end of their confidence intervals. While fluency is influenced by a writer's ability to generate ideas and write without interruption, the substantial interaction effect suggests that task type and task order have a greater impact on fluency than previously assumed. Independent and integrated tasks demand fluency in distinct ways (Barkaoui 2015), and the interplay between task complexity and cognitive load may affect writing fluency more directly than syntactic or lexical complexity measures (Frear and Bitchener 2015). This finding echoes Manchón et al.'s (Manchón et al. 2023) observation that L2 proficiency, not working memory, was the strongest predictor of L2 writing performance under different task conditions. Thus, the learners' ability to fluently navigate varying task demands may hinge more on their linguistic competence and prior task engagement than on cognitive capacity alone.

In contrast, CN/C was not significantly affected by task type, task order, or their interaction. This suggests that certain measures of syntactic complexity, such as the use of complex nominals, may remain relatively stable regardless of task variations (Yang et al. 2015). This stability could be attributed to individual writing styles, proficiency, or other personal factors. Moreover, the nature of integrated tasks, which often emphasize synthesis, may inherently limit the need for complex nominals. Additionally, as Manchón (2020) emphasizes, different task features engage learners in distinct dimensions of writing development. Therefore, the lack of variation in CN/C might reflect the limited sensitivity of this measure to task-based manipulations rather than an absence of learning or strategic adaptation.

7 Conclusion

The study highlights the varying effects of task type and task order on L2 writing performance across the CALF metrics. Task type consistently had a stronger impact than task order, influencing syntactic complexity, lexical complexity, accuracy, and fluency. The INTEG task generally resulted in higher CALF scores than the IND task across all metrics, except for CN/C. While task order interacted with task type to influence performance, its individual effect was limited. Notably, fluency was found to be significantly influenced by both task type and order, as well as their somewhat unexpected interaction. In contrast, CN/C remained unaffected by both task type and order, suggesting that other factors may contribute to its stability.

From a pedagogical perspective, these findings emphasize the importance of task design in L2 writing instruction. Since task type has a more consistent and significant impact on performance, educators should prioritize task selection based on desired learning outcomes. Tasks that encourage complex syntactic structures and varied vocabulary, like the INTEG task, could promote greater linguistic development. Teachers should also consider task sequencing strategically, as the interaction between task order and task type can enhance the effects of specific task types. Incorporating a mix of tasks can help balance different aspects of writing, such as fluency and accuracy, enabling learners to develop a well-rounded skill set.

The study also suggests that task type should be central in TBLT. Tasks promoting syntactic and lexical complexity can challenge learners and enhance higher-level writing skills. Additionally, since task order influences fluency and accuracy, educators should plan task sequences to optimize performance across these areas.

The findings emphasize the need for L2 writing assessment to consider both task type and task order. Given that task type had a significant and consistent impact on performance across multiple metrics, assessment should not solely focus on the final writing outcome but also consider the configuration of tasks involved. Rubrics should be designed to reflect how task type and sequencing influence various aspects of writing, such as syntactic complexity, lexical choice, fluency, and accuracy. Additionally, incorporating different task types and orders into assessments can provide a more nuanced and accurate picture of learners' writing abilities, capturing their strengths and areas for improvement across various writing contexts. This approach acknowledges the complexity of L2 writing performance and ensures a more holistic evaluation of learner progress.

While the findings provide valuable insights, several limitations should be considered. First, the study focused primarily on the comparison between two task types (INTEG and IND), and the results may not generalize to other types of tasks or contexts. Future studies could explore

a wider range of tasks to determine if similar patterns emerge. Additionally, the analysis revealed no significant effects on CN/C, raising questions about why this metric remains unaffected by task type, task order, and their interaction. It may be that other variables, such as task motivation (Dörnyei 2019), cognitive load (Zhang and Zhang 2022), or language mindsets (Lee et al. 2023), are more influential for this particular measure. Further research could explore these additional factors to clarify the dynamics of CN/C.

Another limitation is that the independent and integrated tasks differed not only in task type but also in topic, which may have introduced content-related confounding variables. The independent task prompt on the topic of *teacher respect* and the integrated task prompt on *living on the moon* could have influenced performance, as task content has been shown to affect L2 writing outcomes (Yang et al. 2015). While task type was the primary variable of interest, future studies could control for topic by using parallel prompts or rotating topics across participant groups to better isolate the effects of task type.

Lastly, the relatively short duration of the study limits our understanding of long-term effects. Task effects might vary over time as learners gain more proficiency, which suggests a need for longitudinal studies to explore how task type and order influence writing performance across extended periods. Despite these issues, this study has contributed to enhancing the current understanding of the role of task type, task order, and their interaction concerning independent and integrated writing tasks.

Author Contributions

Mahmoud Abdi Tabari: conceptualization, data collection, writing – original draft, review and editing. **Hansol Lee**: writing – original draft, methodology, software, data curation.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Due to the confidential nature of the data, it cannot be made available to others.

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